

## KROPYVNYSTKYI, Ukraine

WMO: 337110

Lat: 48.5436N

Lon: 32.2833E

Elev: 171

StdP: 99.29

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -18.3      | -14.9 | -21.4                         | 0.6 | -17.7 | -18.0 | 0.8 | -14.4 | 10.5                  | -5.3 | 9.6 | -4.3 | 2.8                   | 330  | 0.560 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.3                   | 32.5            | 19.7 | 30.7 | 19.3 | 29.0 | 18.8 | 21.5                | 28.6 | 20.7 | 27.6 | 19.9 | 26.6 | 4.4                  | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.3                            | 14.4 | 23.9 | 18.5 | 13.7 | 23.2 | 17.7 | 13.0 | 22.5 | 63.4          | 28.6 | 60.5 | 27.6 | 57.8 | 26.5 | 23.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |     |      | Extreme Annual Temperature |       |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |  |
|-------------------|-----|------|----------------------------|-------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|--|
| 1%                |     | 2.5% | 5%                         | Mean  |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |  |
| Min               |     | Max  |                            | Min   | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min  | Max        |      |  |
| (a)               | (b) | (c)  |                            | (e)   | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |
| 9.3               | 8.4 | 7.6  |                            | -22.0 | 34.8 | 3.5                | 2.1                                                | -24.5     | 36.3 | -26.6      | 37.5 | -28.5      | 38.7 | -31.1      | 40.2 |  |
|                   |     |      |                            | -22.2 | 22.6 | 3.4                | 0.8                                                | -24.7     | 23.2 | -26.7      | 23.6 | -28.6      | 24.1 | -31.1      | 24.6 |  |
|                   |     |      |                            | DB    |      |                    |                                                    |           |      |            |      |            |      |            |      |  |
|                   |     |      |                            | WB    |      |                    |                                                    |           |      |            |      |            |      |            |      |  |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 9.1   | -4.1  | -2.7  | 2.3   | 9.9   | 15.7  | 19.3  | 21.5  | 20.8  | 15.3  | 8.8   | 3.0   | -1.9 | (6)  |
| (7)  |                                                                                  | DBStd       | 10.13 | 5.63  | 5.68  | 4.65  | 4.06  | 4.15  | 3.41  | 3.01  | 3.48  | 4.00  | 4.46  | 4.80  | 5.37 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1751  | 437   | 356   | 239   | 49    | 6     | 0     | 0     | 0     | 5     | 77    | 213   | 368  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 3674  | 695   | 589   | 497   | 252   | 102   | 29    | 7     | 14    | 106   | 298   | 459   | 627  | (9)  |
| (10) |                                                                                  | CDD10.0     | 1409  | 0     | 0     | 2     | 48    | 183   | 279   | 356   | 334   | 165   | 39    | 4     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 290   | 0     | 0     | 0     | 1     | 21    | 58    | 104   | 89    | 16    | 1     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 3026  | 0     | 0     | 0     | 14    | 239   | 575   | 1023  | 987   | 182   | 6     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 1017  | 0     | 0     | 0     | 2     | 55    | 163   | 357   | 393   | 47    | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.8   | 4.1   | 4.2   | 4.4   | 4.1   | 3.6   | 3.4   | 3.2   | 3.3   | 3.4   | 3.5   | 4.0   | 4.0  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 499   | 32    | 29    | 27    | 35    | 46    | 66    | 64    | 47    | 39    | 28    | 37    | 40   | (15) |
| (16) |                                                                                  | PrecMax     | 751   | 129   | 68    | 67    | 79    | 102   | 216   | 150   | 131   | 179   | 106   | 156   | 97   | (16) |
| (17) |                                                                                  | PrecMin     | 328   | 8     | 1     | 3     | 3     | 2     | 9     | 1     | 0     | 0     | 0     | 4     | 4    | (17) |
| (18) |                                                                                  | PrecStd     | 98    | 24    | 16    | 17    | 19    | 19    | 44    | 36    | 31    | 36    | 25    | 31    | 23   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 7.5   | 9.8   | 17.2  | 25.0  | 31.0  | 32.2  | 34.3  | 35.6  | 30.8  | 23.6  | 16.4  | 10.3 | (19) |
| (20) |                                                                                  | 0.4%        | MCWB  | 6.3   | 7.0   | 10.0  | 13.8  | 17.6  | 19.6  | 20.6  | 20.0  | 17.5  | 15.2  | 12.7  | 9.2  | (20) |
| (21) |                                                                                  | 2%          | DB    | 5.4   | 7.2   | 14.1  | 21.8  | 28.0  | 30.2  | 32.2  | 33.2  | 27.8  | 20.6  | 13.2  | 8.3  | (21) |
| (22) |                                                                                  | 2%          | MCWB  | 4.3   | 5.5   | 8.3   | 12.6  | 16.9  | 19.1  | 20.1  | 19.6  | 16.7  | 14.2  | 11.2  | 7.3  | (22) |
| (23) |                                                                                  | 5%          | DB    | 3.7   | 5.4   | 11.3  | 19.7  | 26.1  | 28.6  | 30.4  | 30.8  | 25.3  | 17.7  | 11.3  | 6.6  | (23) |
| (24) |                                                                                  | 5%          | MCWB  | 2.9   | 4.1   | 6.8   | 11.5  | 16.5  | 18.7  | 19.9  | 19.2  | 15.8  | 13.2  | 10.0  | 5.8  | (24) |
| (25) |                                                                                  | 10%         | DB    | 2.4   | 3.8   | 9.0   | 17.5  | 23.8  | 26.9  | 28.6  | 28.7  | 22.9  | 15.4  | 9.6   | 4.8  | (25) |
| (26) |                                                                                  | 10%         | MCWB  | 1.8   | 2.6   | 5.6   | 10.6  | 15.6  | 18.3  | 19.4  | 18.5  | 15.1  | 12.1  | 8.5   | 4.2  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 6.6   | 7.6   | 10.6  | 15.4  | 19.5  | 21.9  | 22.6  | 22.2  | 19.6  | 17.4  | 13.4  | 9.6  | (27) |
| (28) |                                                                                  | 0.4%        | MCDWB | 7.3   | 9.1   | 16.2  | 23.8  | 26.7  | 28.3  | 29.5  | 30.2  | 27.2  | 21.3  | 15.4  | 10.2 | (28) |
| (29) |                                                                                  | 2%          | WB    | 4.4   | 5.9   | 8.7   | 13.6  | 18.2  | 20.8  | 21.7  | 21.2  | 18.0  | 15.3  | 11.6  | 7.4  | (29) |
| (30) |                                                                                  | 2%          | MCDWB | 5.1   | 6.9   | 13.2  | 19.6  | 25.2  | 27.1  | 28.5  | 28.8  | 24.6  | 18.2  | 13.1  | 8.0  | (30) |
| (31) |                                                                                  | 5%          | WB    | 3.0   | 4.2   | 7.2   | 12.4  | 17.2  | 19.9  | 21.0  | 20.2  | 16.8  | 13.8  | 10.1  | 5.9  | (31) |
| (32) |                                                                                  | 5%          | MCDWB | 3.6   | 5.3   | 10.4  | 17.6  | 24.0  | 26.1  | 27.9  | 27.7  | 22.7  | 17.0  | 11.1  | 6.5  | (32) |
| (33) |                                                                                  | 10%         | WB    | 1.9   | 2.8   | 5.9   | 11.4  | 16.2  | 19.0  | 20.2  | 19.3  | 15.9  | 12.4  | 8.7   | 4.2  | (33) |
| (34) |                                                                                  | 10%         | MCDWB | 2.4   | 3.7   | 8.5   | 16.2  | 22.5  | 24.7  | 26.6  | 26.4  | 21.4  | 15.2  | 9.5   | 4.7  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               |             | MDBR  | 4.8   | 5.6   | 7.5   | 10.2  | 11.1  | 10.7  | 11.3  | 11.9  | 10.4  | 8.4   | 5.4   | 4.7  | (35) |
| (36) |                                                                                  | 5% DB       | MCDBR | 4.7   | 6.9   | 11.8  | 13.7  | 14.1  | 13.4  | 13.8  | 15.2  | 14.6  | 12.0  | 7.9   | 5.7  | (36) |
| (37) |                                                                                  |             | MCWBR | 4.0   | 5.4   | 7.1   | 6.4   | 5.7   | 5.0   | 4.6   | 4.8   | 5.9   | 6.4   | 5.8   | 4.8  | (37) |
| (38) |                                                                                  |             | MCDBR | 4.3   | 6.6   | 10.6  | 11.6  | 12.1  | 11.2  | 11.7  | 12.7  | 12.0  | 10.1  | 6.9   | 5.3  | (38) |
| (39) |                                                                                  | 5% WB       | MCWBR | 3.9   | 5.4   | 6.8   | 5.8   | 5.5   | 4.9   | 4.4   | 4.6   | 5.7   | 6.3   | 5.6   | 4.7  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.321 | 0.352 | 0.383 | 0.442 | 0.414 | 0.406 | 0.437 | 0.432 | 0.404 | 0.367 | 0.349 | 0.324 | (40) |      |
| (41) |                                                                                  | taud        | 2.231 | 2.182 | 2.215 | 2.101 | 2.267 | 2.304 | 2.223 | 2.229 | 2.278 | 2.365 | 2.327 | 2.303 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 717   | 778   | 820   | 805   | 849   | 856   | 822   | 807   | 791   | 762   | 684   | 666   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 78    | 103   | 118   | 147   | 129   | 125   | 134   | 128   | 110   | 86    | 72    | 65    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.96  | 1.81  | 3.03  | 4.52  | 5.81  | 6.24  | 6.12  | 5.49  | 3.76  | 2.26  | 0.95  | 0.67  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.18  | 0.34  | 0.48  | 0.42  | 0.48  | 0.47  | 0.34  | 0.33  | 0.53  | 0.32  | 0.16  | 0.14  | (45) |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | +0.69 | N/A     | N/A    | +0.99   | N/A   | N/A   | -145        | -192    | +110    | +68     |
| (47) | Regional (9 neighbors) | +0.72 | N/A     | N/A    | +1.01   | N/A   | N/A   | -140        | -201    | +129    | +80     |

Nomenclature: See separate page